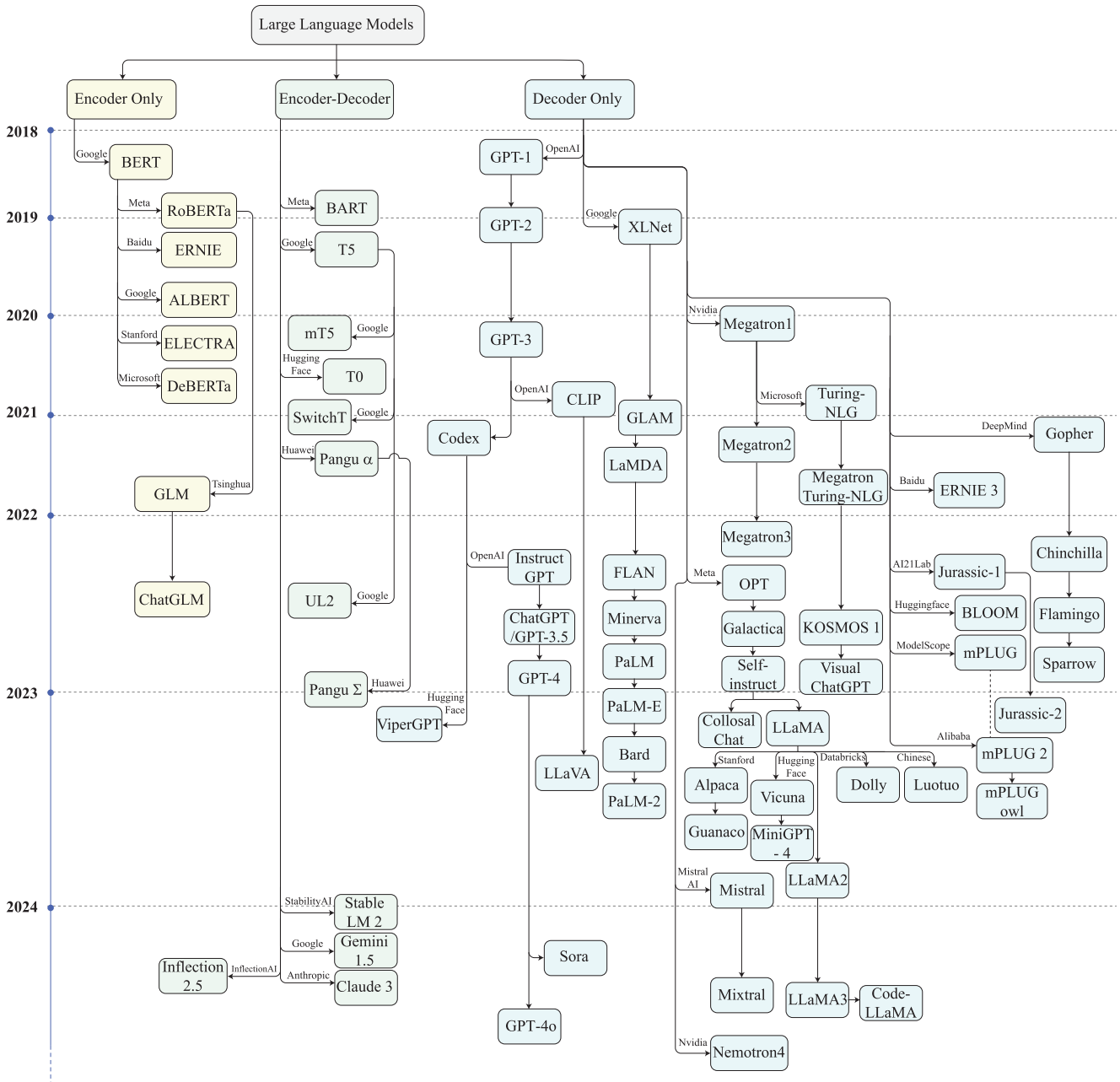


FIGURE 12. An evolutionary tree illustrating the progression of mainstream LLMs. Models are categorized based on their architectural construct: encoder-only, decoder-only, and encoder-decoder. Models stemming from the same lineage or released by identical entities are interconnected with solid lines. Independent research contributions are demarcated by purple lines.

Figure 14 displays the average scores from the Open-LLM Leaderboard for various benchmarks, including MMLU, ARC, HellaSwag, and TruthfulQA.

A. AUTO-ENCODING MODELS

Auto-encoding models, often referred to as 'encoder-only models,' utilize solely the encoder component of the Transformer architecture. Their primary function is to map input data from a higher-dimensional space to a lower-dimensional vector space, effectively capturing and integrating contextual information into the data representation. These models commonly employ training strategies like masked language modeling and bi-directional training. The inception of auto-encoding models can be traced back to 2018 with the release of BERT, a pioneering model that harnessed the encoder-only architecture. This innovation significantly augmented the capabilities of natural language understanding models, enabling them to tackle a diverse range of NLU tasks, including reading comprehension, cloze tasks, and question answering.

1) BERT

BERT [9] stands out as one of the pioneering pre-trained models employing an auto-encoding architecture. Its training